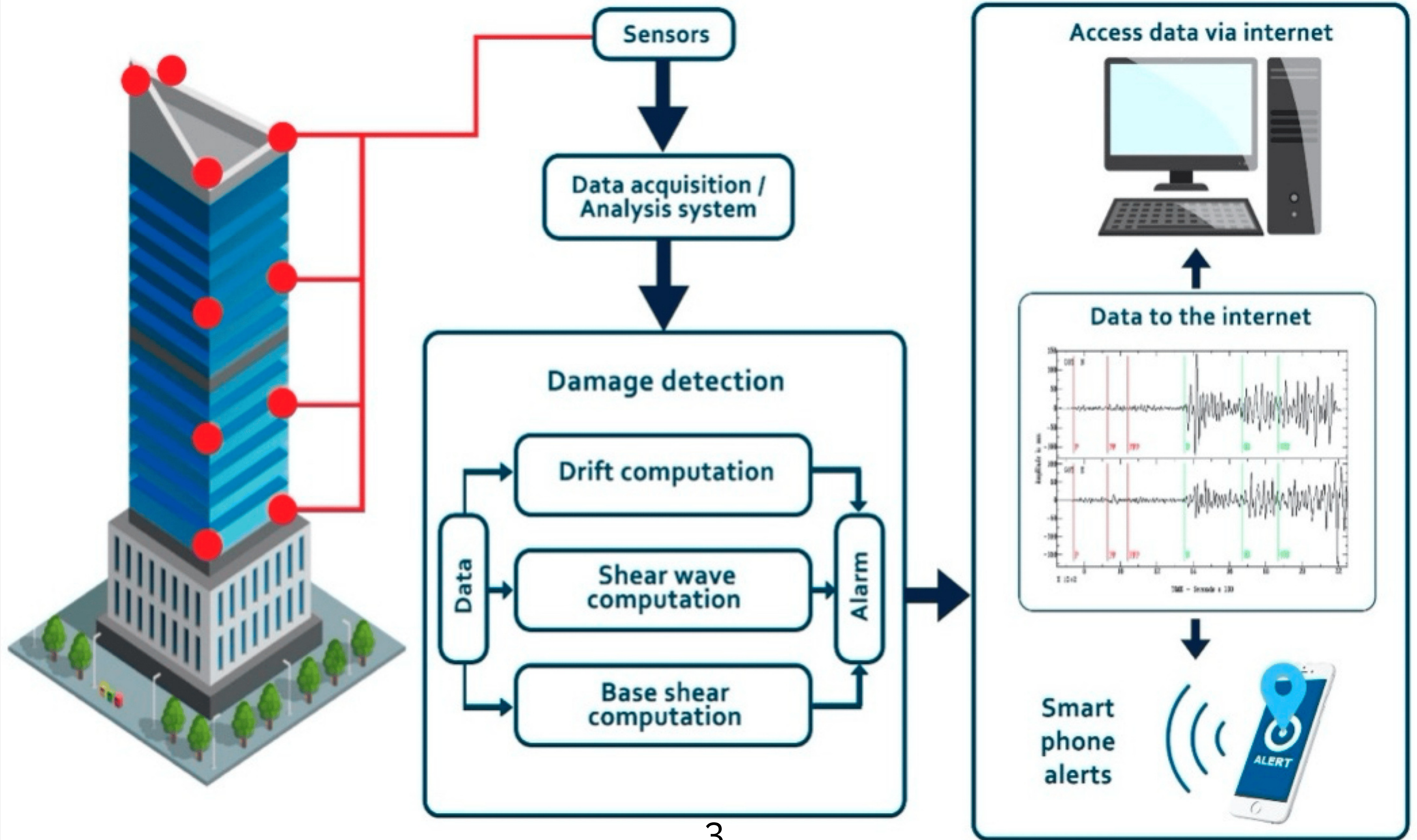


Structural Health Monitoring Internship Program

**With
paritta shm in association
with
VJTI tbi and structural engineering
department**





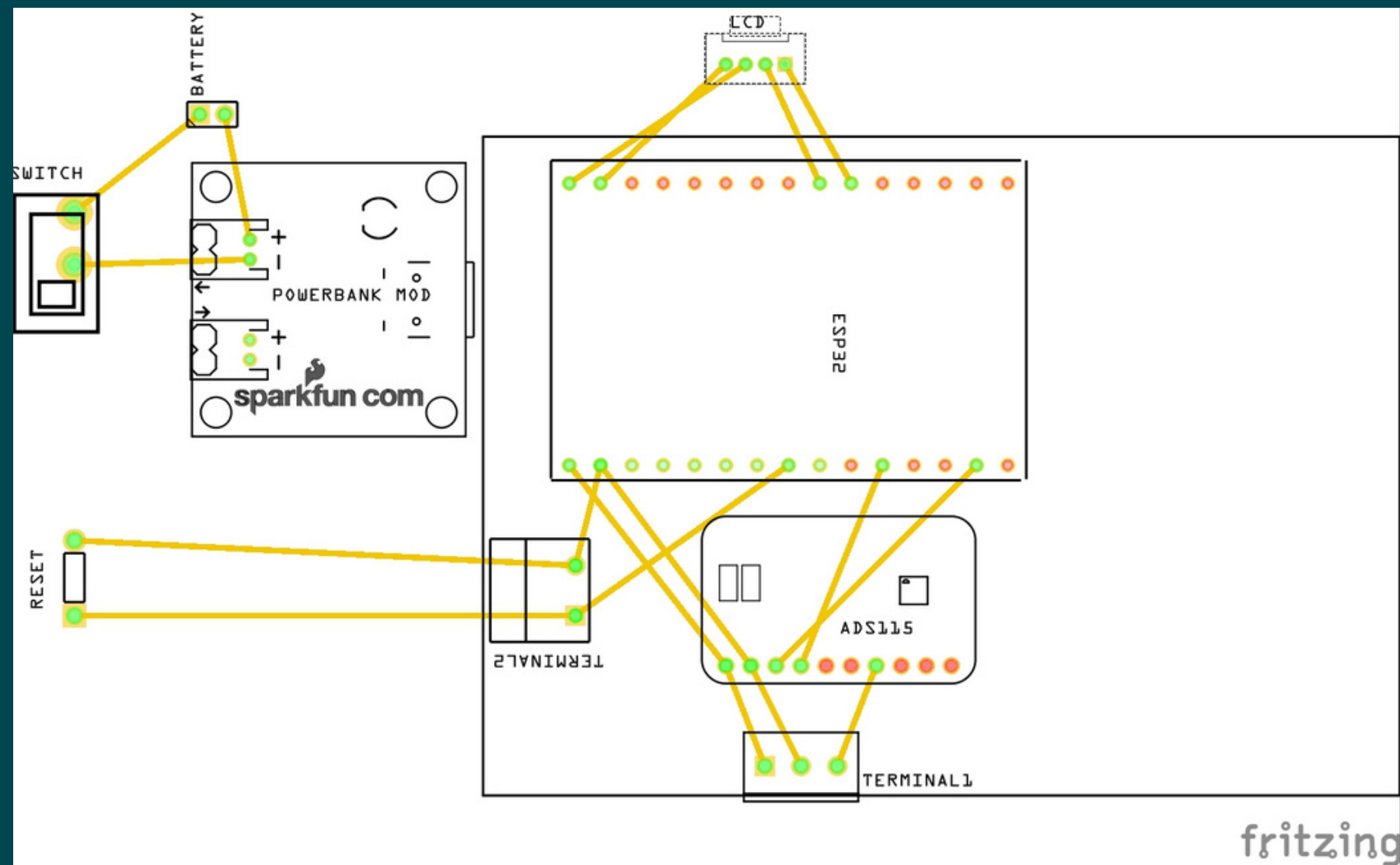
IoT based Potentiometer

A crack alert system using a potentiometer designed to monitor the structural health of a particular component or structure and provide an alert when cracks or damages are detected. The potentiometer acts as a sensor to measure changes in the structure's deformation or strain, which can occur due to cracks or damage.

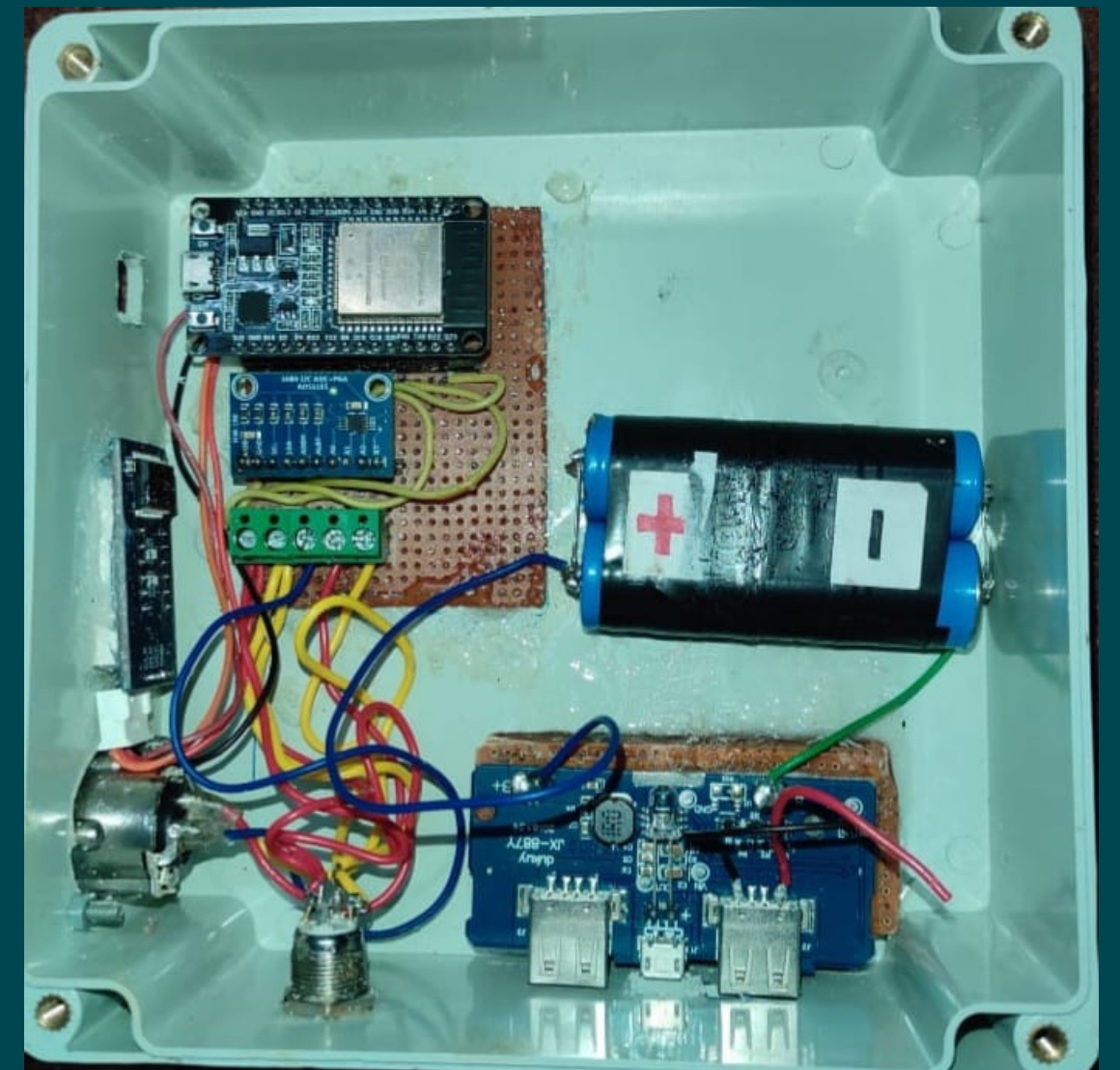
Following are buyers of such technologies and services :

1. Engineering and Construction Firms
2. Infrastructure Owners and Operators
3. Transportation Industry
4. Energy and Utility Companies
5. Government Agencies

Circuit Diagram



Operational Circuit



➤ Sensor Used :

- elap Linear Potentiometer Series PM

What is Potentiometer ?

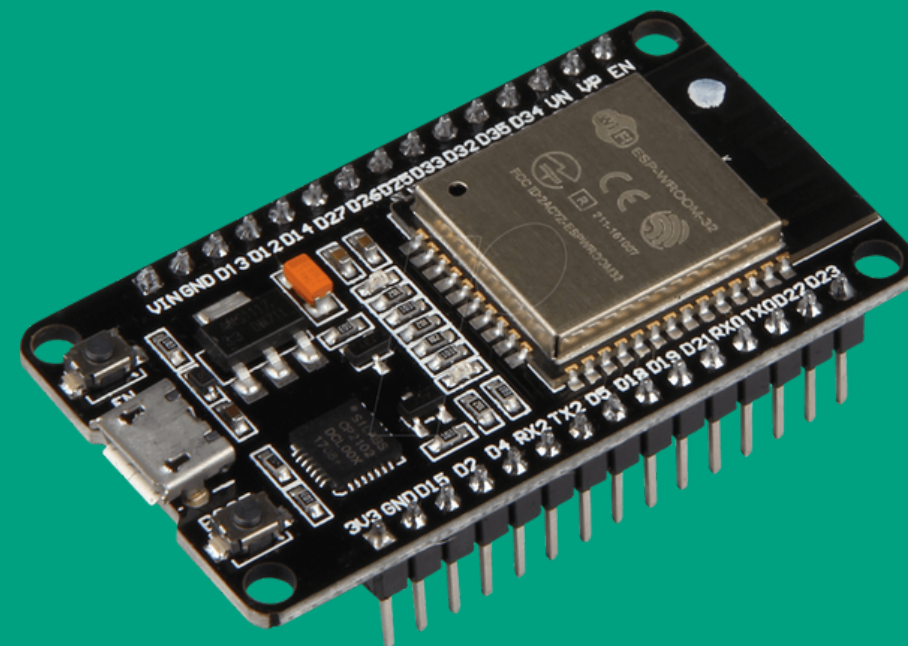
A potentiometer (also known as a pot or potmeter) is defined as a 3 terminal variable resistor in which the resistance is manually varied to control the flow of electric current.



Components Used :

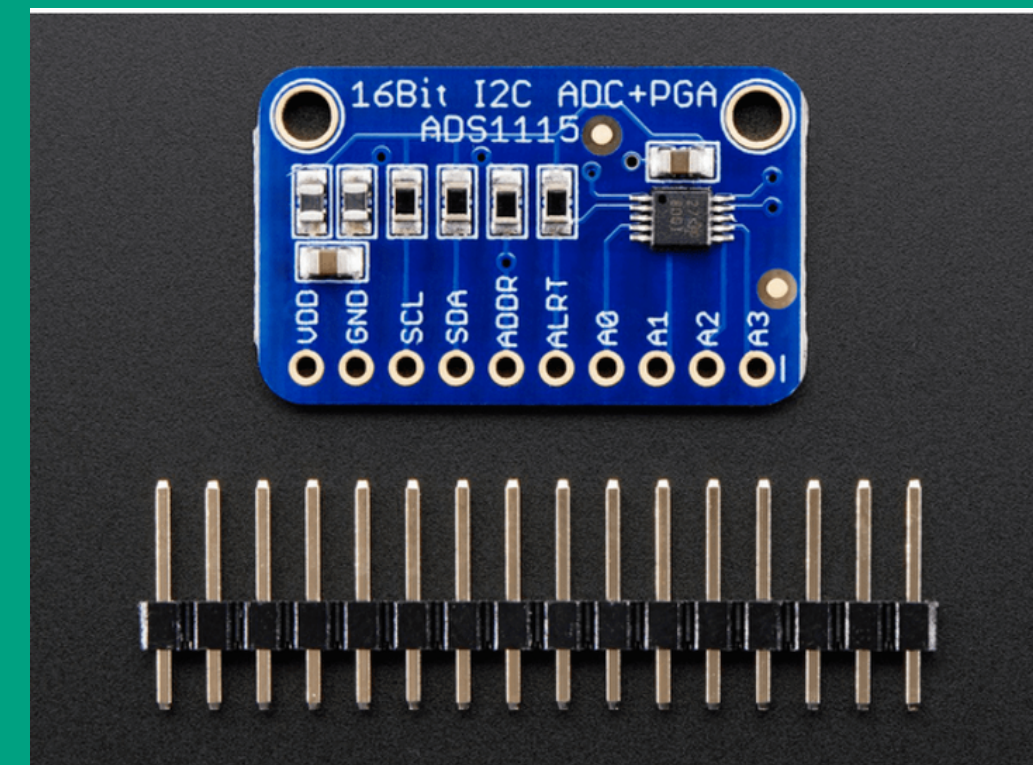
ESP-WROOM-32

- It has onboard 18 Analog to digital converts ADCs.
- Resolution of each ADCs is configurable from 9 - 12 bit.
- Faster Wi-Fi.
- Supports Bluetooth 4.2 and Bluetooth Low Energy.



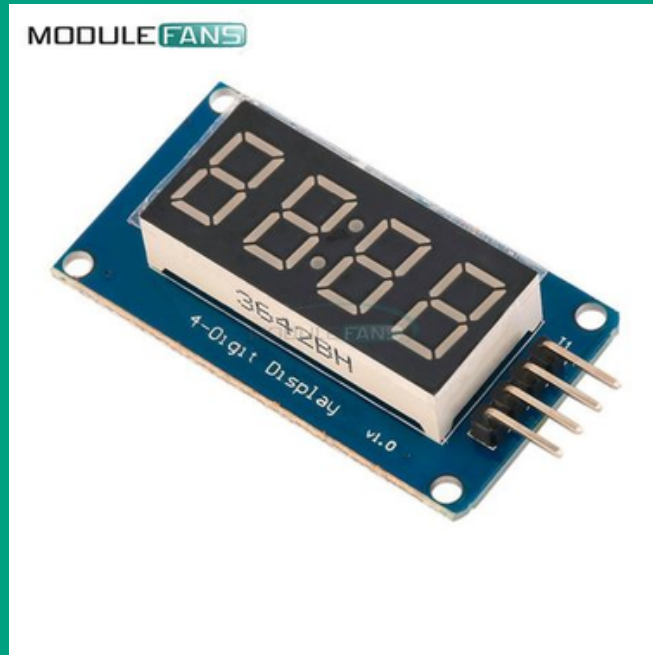
16 Bit I2C ADs 1115

- Is accurate to 0.01% .
- It has a low power consumption.
- It amplifies the resolution to 16 bits.



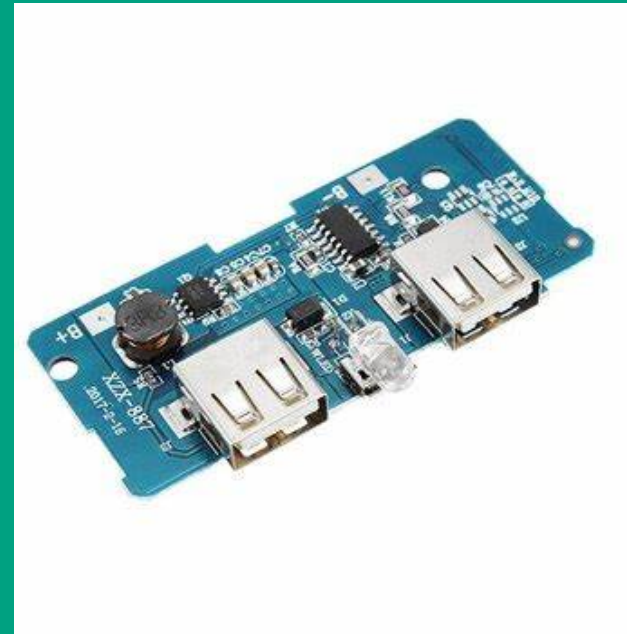
TM1637 4-Digit 7-Segment Display

- Easy to install.
- Cost effective.



Dual USB 5V Power bank module

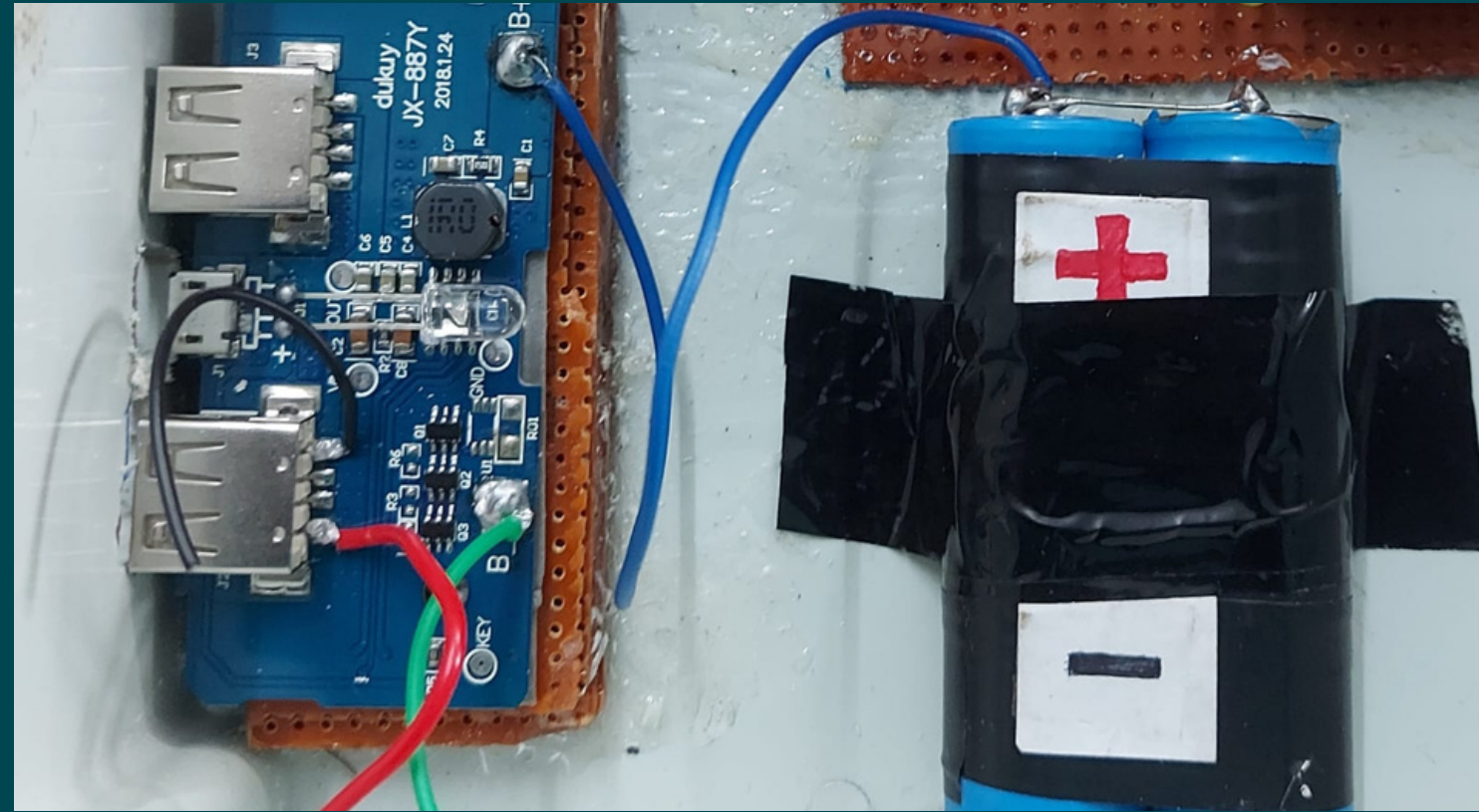
- Overdischarge protection.
- Over temperature protection.
- Short circuit protection.



18650 Li-ion 2000mAh rechargeable cells

- low internal resistance.
- long battery life.
- It can be charged and discharged up to 1000 cycles without much loss in battery capacity.





- Power Bank Module(5V) with LiPo Battery
- LiPo battery is rechargable with 2000 mAh Capacity.
- We can change circuit with inbuilt port of power bank of module



TESTING

- 1 . IOT based Crack Measurement device**
- 2 . Crack Width meter (Analog Device)**

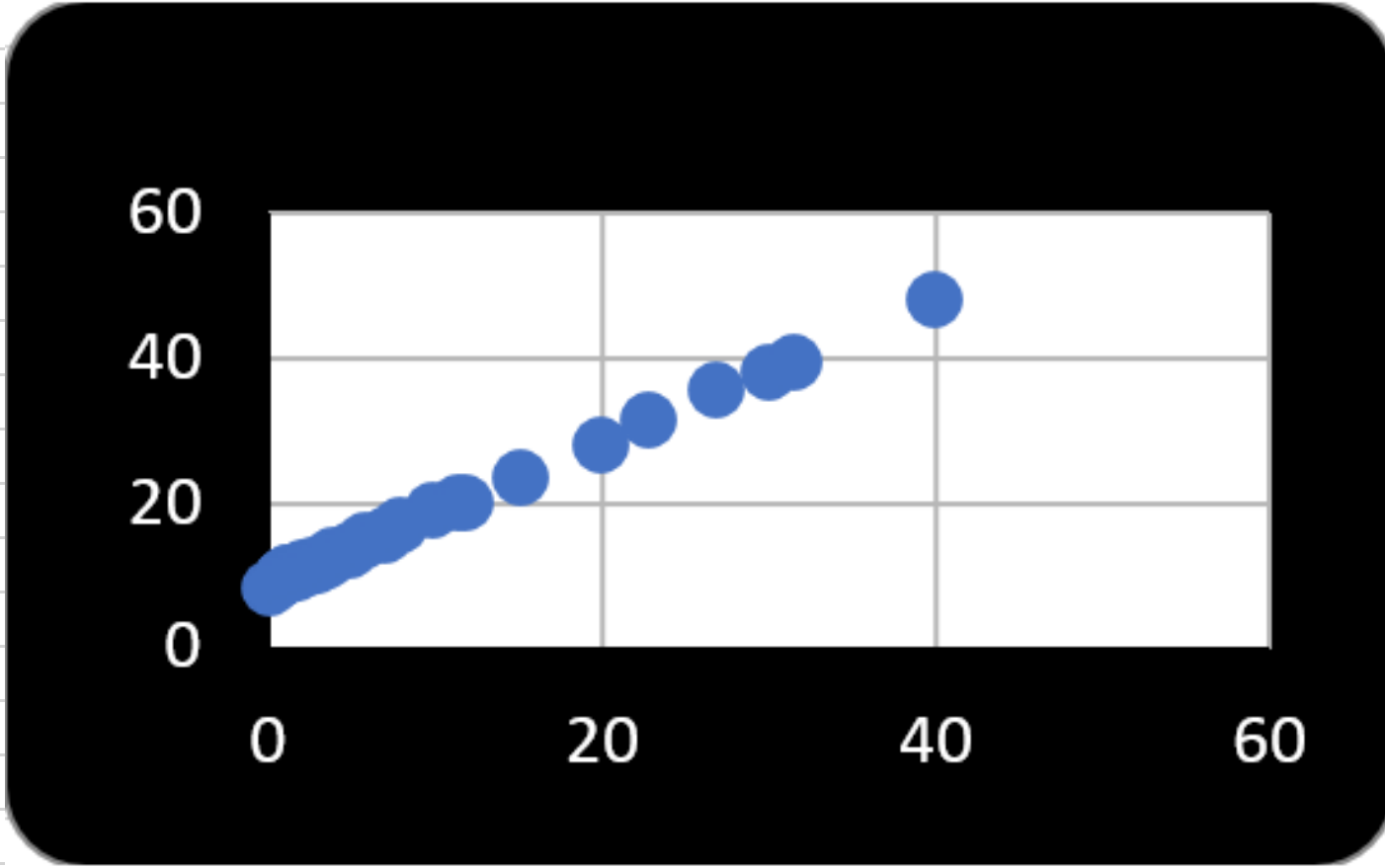
Testing and Accuracy Check

Gauge blocks are a system for producing precision lengths. The individual gauge block is a metal or ceramic block that has been precision ground and lapped to a specific thickness.

Wringing is the process of sliding two blocks together so that their faces bond. Because of their ultraflat surfaces, when wrung, gauge blocks adhere to each other tightly.

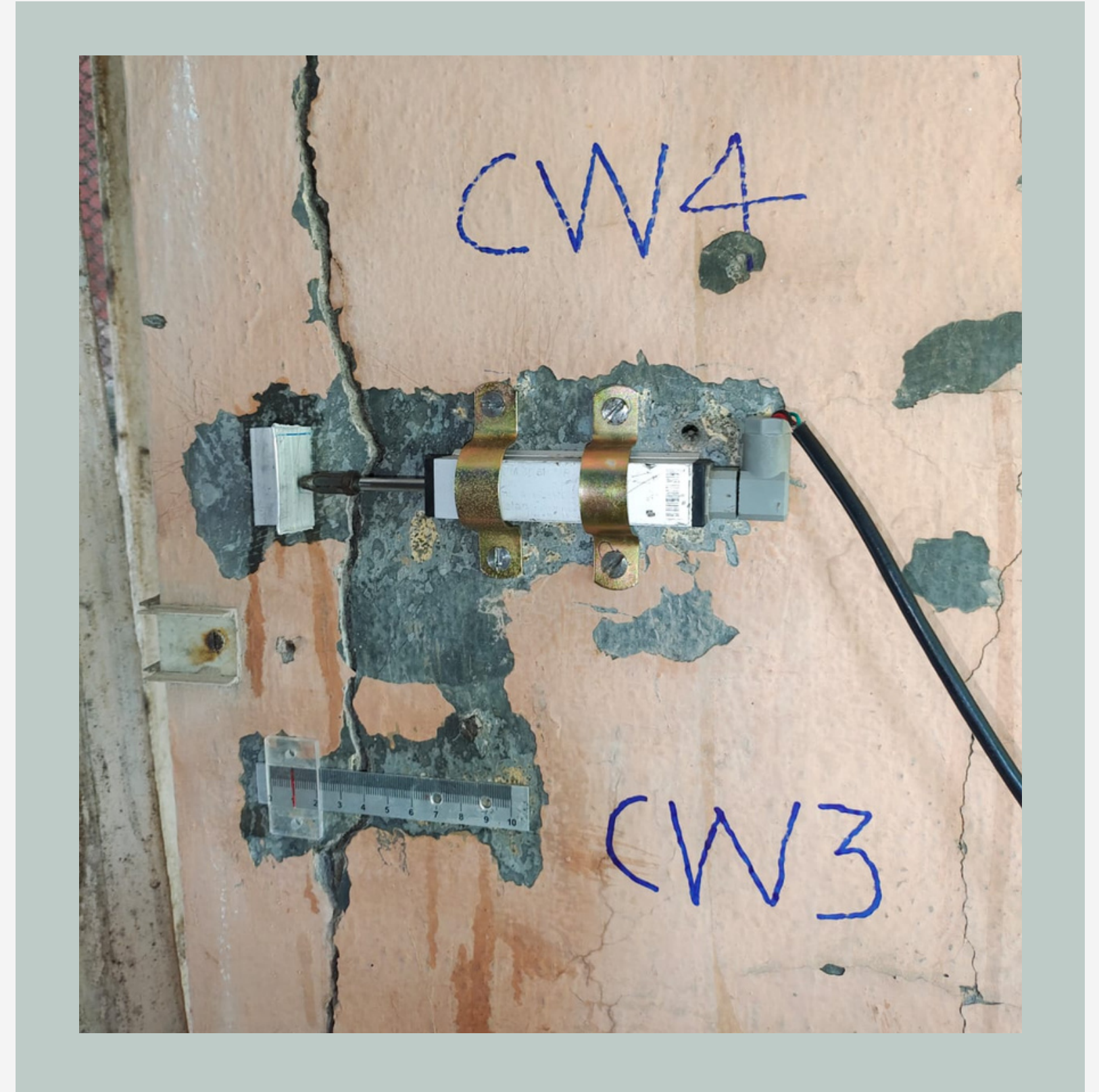


no	gage block reading (mm)	Circuit Reading (mm)	Difference	Absolute difference	Relative error
1	0	8.08	0	0	0
2	1	9.09	1.01	0.01	0.01
3	1.01	9.12	1.04	0.03	0.02970297
4	1.02	9.17	1.09	0.07	0.068627451
5	1.03	9.21	1.13	0.1	0.097087379
6	1.05	9.22	1.14	0.09	0.085714286
7	1.07	9.23	1.15	0.08	0.074766355
8	1.09	9.29	1.21	0.12	0.110091743
9	1.1	9.29	1.21	0.11	0.1
10	1.3	9.5	1.42	0.12	0.092307692
11	1.4	9.61	1.53	0.13	0.092857143
12	1.5	9.72	1.64	0.14	0.093333333
13	1.8	10.03	1.95	0.15	0.083333333
14	2	10.21	2.13	0.13	0.065
15	2.3	10.47	2.39	0.09	0.039130435
16	2.8	10.94	2.86	0.06	0.021428571
17	3	11.17	3.09	0.09	0.03
18	3.5	11.74	3.66	0.16	0.045714286
19	4	12.33	4.25	0.25	0.0625
20	5	13.27	5.19	0.19	0.038
21	6	14.27	6.19	0.19	0.031666667
22	7	15.27	7.19	0.19	0.027142857
23	8	16.23	8.15	0.15	0.01875
24	10	18.19	10.11	0.11	0.011
25	11.5	19.65	11.57	0.07	0.006086957
26	12	20.1	12.02	0.02	0.001666667
27	15	23.04	14.96	-0.04	0.002666667
28	20	27.93	19.85	-0.15	0.0075
29	23	30.89	22.81	-0.19	0.00826087
30	27	34.8	26.72	-0.28	0.01037037
31	30	37.81	29.73	-0.27	0.009
32	31.5	39.27	31.19	-0.31	0.00984127
33	40	47.62	39.54	-0.46	0.0115
					SUM= 1.395047301
			accuracy :-	0.042274161	



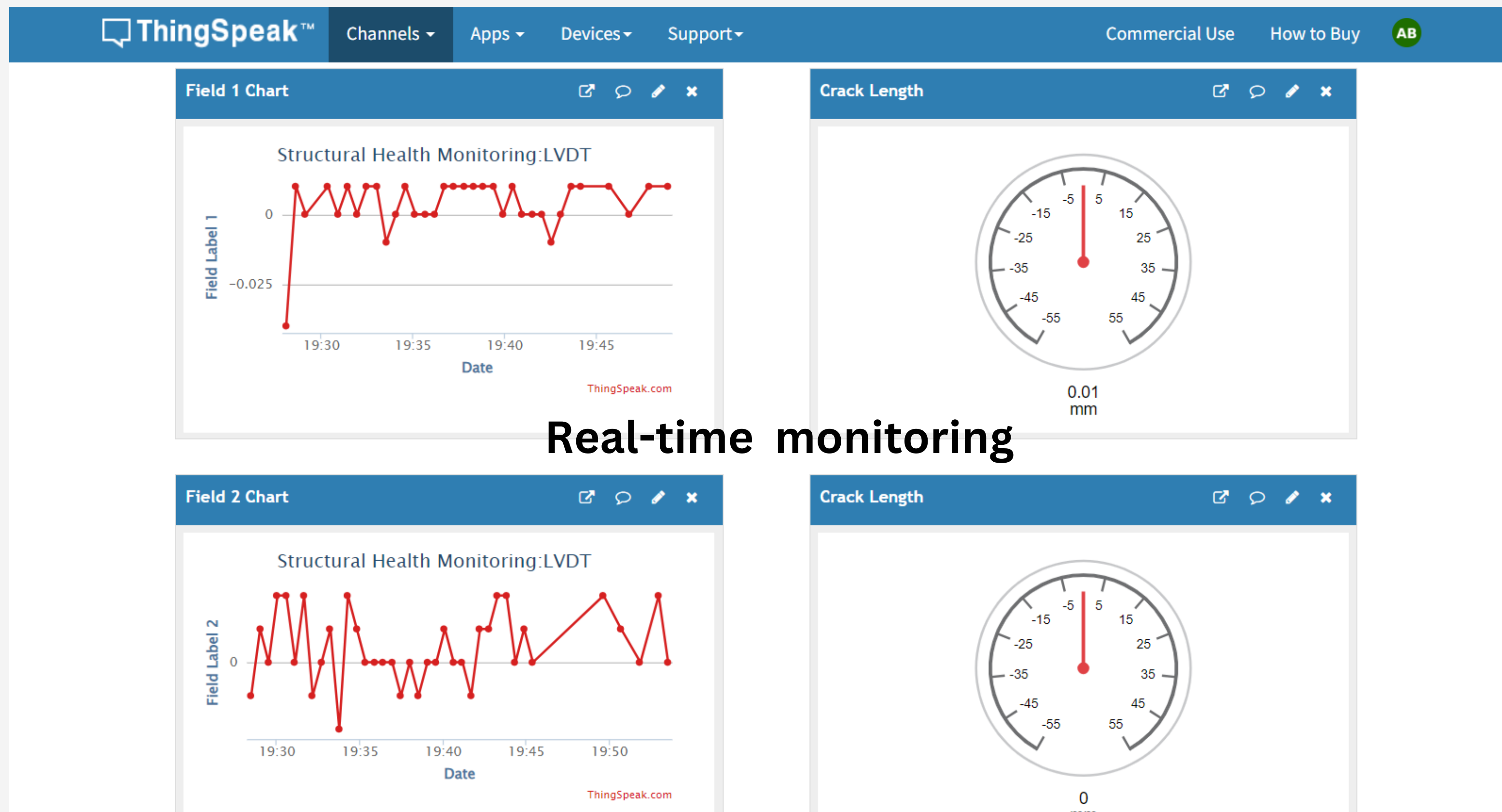


Setup :-





RESULTS :-



Conclusion :-

- Based on the results, our IoT-based crack measurement device outperformed the traditional analog method in terms of accuracy and efficiency.
- The device's ability to detect smaller cracks early on can significantly enhance structural safety and prevent potential damages.

*Thank
you!*